Bulk Certificate Generator

Test Execution Instructions and Expected Results

Purpose

This guide explains how to run the project's automated tests and what each test checks. The suite covers job creation, request validation, PDF generation, job status and progress, per-recipient failure isolation, and certificate downloads.

The tests use FastAPI's TestClient with an isolated in-memory SQLite database and temporary file storage. You do not need to start the web server or use the Swagger page to run them. The manual scenario command separately creates persistent example PDFs and ZIP bundles.

1. Prepare PowerShell

Open PowerShell and run the following commands from the project root:

```
cd C:\Users\SAMVAADI\bulk-certificate-generator
python -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install -r requirements.txt
```

If PowerShell blocks activation, allow scripts only in the current PowerShell process, then activate the environment:

```
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
.\.venv\Scripts\Activate.ps1
```

A successful activation normally prefixes the command prompt with (.venv). Use Python 3.11 or later.

2. Run the complete automated suite

```
python -m pytest -q
```

Expected result: all tests pass. The verified project currently reports 7 passed. A Starlette/httpx deprecation warning may be printed by the installed dependency versions; it does not indicate a test failure.

Test Cases and Inputs

The commands below run one test at a time. Execute them from the project root with the virtual environment active.

A. Create a generation job

Command: python -m pytest -q tests/test_jobs.py::test_create_job_returns_202_and_finishes
Input: Python Workshop; issue date 2026-10-09; three fictional recipients (Ada Lovelace, Grace Hopper, Alan Turing) with valid example.com email addresses. Ada also has metadata grade=A.

Expected: POST /api/v1/jobs returns HTTP 202 and PENDING with total_recipients=3. The background task completes during the test request; the subsequent status request reports COMPLETED and progress_percentage=100.

B. Validate input

Command: python -m pytest -q tests/test_jobs.py::test_invalid_payloads_return_422
Inputs: an empty recipients array; a malformed email address; and an empty recipient name.

Expected: each invalid request returns HTTP 422 and is rejected before a generation job is accepted.

C. Generate a certificate PDF

Command: python -m pytest -q tests/test_certificates.py::test_certificate_generation_returns_pdf_bytes
Input: Test Event, issue date 2026-10-10, Taylor Example, a verification ID, and metadata grade=A / issuer=Test Organization.

Expected: the generator returns non-empty bytes beginning with the PDF signature %PDF-.

D. Check job status and progress

Command: python -m pytest -q tests/test_jobs.py::test_create_job_returns_202_and_finishes
Input: the same three-recipient request from case A.

Expected: status polling reports COMPLETED, completed_count=3, failed_count=0, and progress_percentage=100.

E. Isolate an individual rendering failure

Command: python -m pytest -q tests/test_jobs.py::test_failure_is_isolated
Input: the three-recipient request from case A. A test renderer deliberately raises ValueError('render failed') for Grace Hopper and returns PDF bytes for Ada and Alan.

Expected: remaining recipients succeed; the job is PARTIALLY_COMPLETED with completed_count=2, failed_count=1, progress_percentage=100, and the failed recipient has error_message='render failed'.

Test Cases and Outputs

F. Retrieve generated certificates

Command: python -m pytest -q tests/test_certificates.py::test_individual_and_bulk_download
Input: the three-recipient Python Workshop job. The test takes a successful recipient's download_url and the job ID from the create/status responses.

Expected: the individual download returns HTTP 200 with application/pdf. The bulk download returns HTTP 200 with application/zip. The separate not-found test checks that an unknown recipient and unknown job return HTTP 404:

```
python -m pytest -q tests/test_certificates.py::test_missing_certificate_and_job
```

G. Run the manual batch scenarios

```
Command: python -m tests.manual_scenarios
```

This command runs end-to-end TestClient scenarios using fictional data and saves the generated archives under storage\certificates\. It verifies a one-recipient job, a five-recipient job with one injected rendering failure, a 100-recipient job, and a 1,000-recipient capacity probe.

Scenario	Expected job result	Saved ZIP
1 recipient	COMPLETED; 1 success, 0 failures	scenario_1_recipient.zip
5 recipients, 1 render failure	PARTIALLY_COMPLETED; 4 successes, 1 failure	scenario_5_one_failure.zip
100 recipients	COMPLETED; 100 successes	scenario_100_recipients.zip
1,000 recipient probe	COMPLETED; 1,000 successes	capacity_probe_1000.zip

The API currently has no configured maximum batch size. The 1,000-recipient scenario is the largest batch verified by this runner, not a guaranteed upper limit. Scenario job records use an in-memory database and disappear when the runner exits; the generated PDF files and ZIPs remain on disk.

3. Inspect outputs and troubleshoot

Open a saved archive in File Explorer by pasting this path into its address bar: C:\Users\SAMVAADI\bulk-certificate-generator\storage\certificates\. The individual PDF files are in subfolders named by job ID.

If pytest reports 'No module named pytest', activate the project's .venv and install requirements again. If a test fails, retain the full pytest output: it identifies the failing test and assertion. A nonzero pytest exit code means the run did not pass.

Detailed test coverage is implemented in tests/test_jobs.py, tests/test_certificates.py, tests/conftest.py, and tests/manual_scenarios.py.